Appendix for D-DietMMR: A Multi-Modal Reasoning Benchmark for Disease-Oriented Diet Recommendations in LVLMs

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1 Annotation Consistency Results

To evaluate annotation quality, we conduct a consistency experiment. We focus on the second stage of the three-stage annotation process (*Second-round checking*) for two reasons: (1) Two annotators independently review the same data, allowing direct measurement of consistency; (2) The task is a binary classification, making evaluation straightforward. We use Cohen's Kappa to measure agreement between annotators. The result is 0.95, indicating high annotation quality.

2 Review Annotation Tool and Compensation

To enhance review annotation efficiency, we develop a tool used for reviewing and checking samples in D-DietMMR.

In our study, the Review Annotation Tool involves a manual checking stage and a manual reviewing stage. During the checking stage, two annotators are responsible for verifying whether the ingredient annotation granularity meets the required standards (the interface for which is shown in Figure A1). The other two annotators check the recommendations of dish-disease samples, as illustrated in Figure A2. For the reviewing stage, the corresponding page is shown in Figure A3.

We compensate annotators at RMB 50 per hour, exceeding the local minimum wage in China. Considering the nature, this rate is fair and competitive.

3 Baselines

To establish the baseline performance in the D-DietMMR test set, we experiment with 18 representative LVLMs with varying parameter sizes under three common prompting strategies. Specifically, four groups of model are considered: For **closed-source LVLMs**, we select 5 models comprising GPT-4o[1], GPT-4v [1], Qwen-VL-Max [2], GLM-4-Plus[2], and Gemini-Pro-V [9]. For **open-source LVLMs**, we choose models from the LLaVA [6], LLama3.2 series[3], Qwen2-VL series [10], and InternVL2 series[4]. For **medical special LVLMs**, we employ HuatuoGPT-V-7B [3], LLaVA-Med-7B [4], HealthGPT-M3 [5], and MedGemma series [8]. For **other methods**, we compare with CILP [7], and BLIP2-Chinese[5].

[1] https://openai.com/index/gpt-4o-system-card/
[2] https://bigmodel.cn/dev/howuse/glm-4v

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4 Group analysis for dish and disease types.

To demonstrate the evaluation performance of D-DietMMR on various LVLMs, we conduct comparison experiments focusing on dishes and diseases. The experiments involve grouping the top-5 types based on sample count from the test set. Top-5 dishes: **Sla.** Slack pepper beef; **Ste.** Stewed chicken; **Don.** Dongpo knuckle; **Sti.** Stir-fried pork liver; **Whi.** White gourd soup. Top-5 diseases: **Rhi.** Rhinitis; **Dia.** Diabetes; **Ora.** for Oralulcer; **Adr.** Adrenocortical carcinoma; **Pne.** Pneumoconiosis. The results, shown in Table A1, reveal significant performance differences across types, indicating the high sample diversity and complexity of the D-DietMMR benchmark.

Similarly, we consider three prompting strategies: **Zero-shot Direct Prompting**, where the model directly outputs the answer, as shown in Figure A4; **One-shot Prompting**, where the LVLM is given an example to guide the output, as shown in Figure A5; and **Chain-of-Thought (CoT) Prompting**, where the LVLM first generates step-by-step reasoning and then outputs the answer, as shown in Figure A6.

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[4] https://internvl.github.io/blog/2024-07-02-InternVL-2.0/
[5] https://www.modelscope.cn/models/xiajinpeng123/BLIP2-Chinese

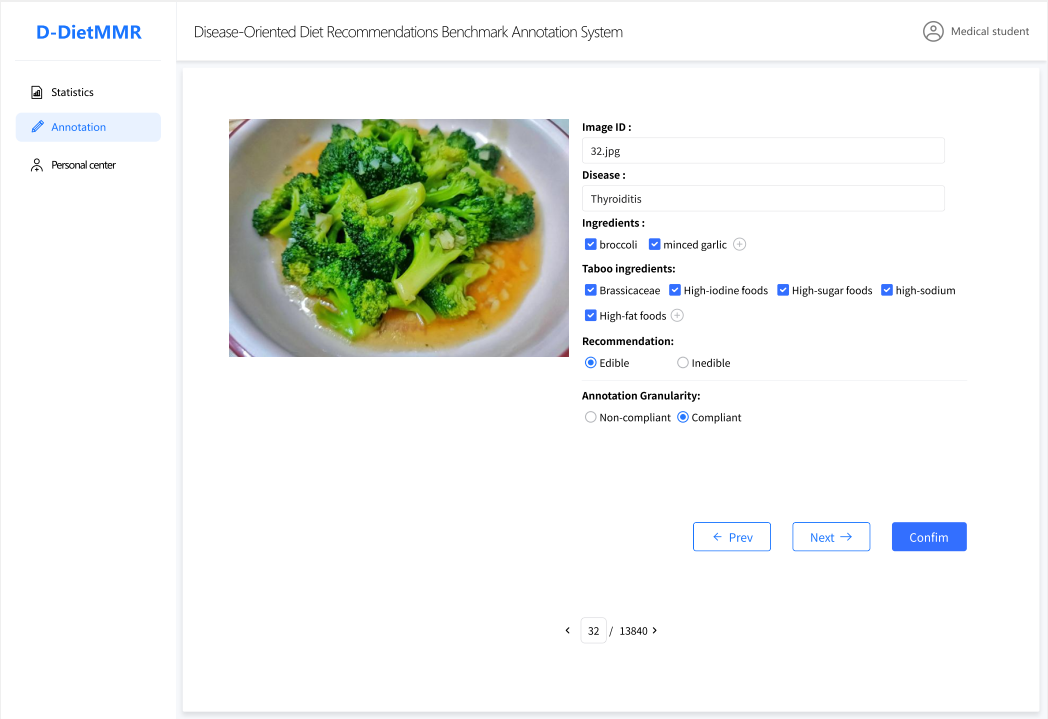

Figure A1: The checking page of annotation granularity in the review annotation tool.

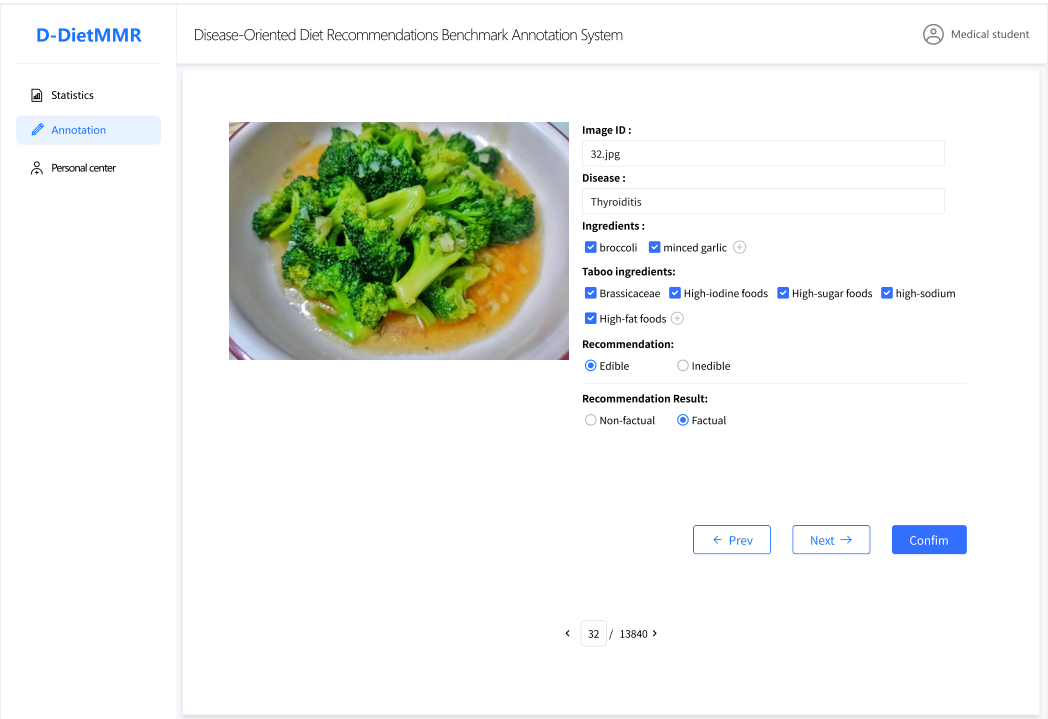

Figure A2: The checking page of the recommendation results in the review annotation tool.

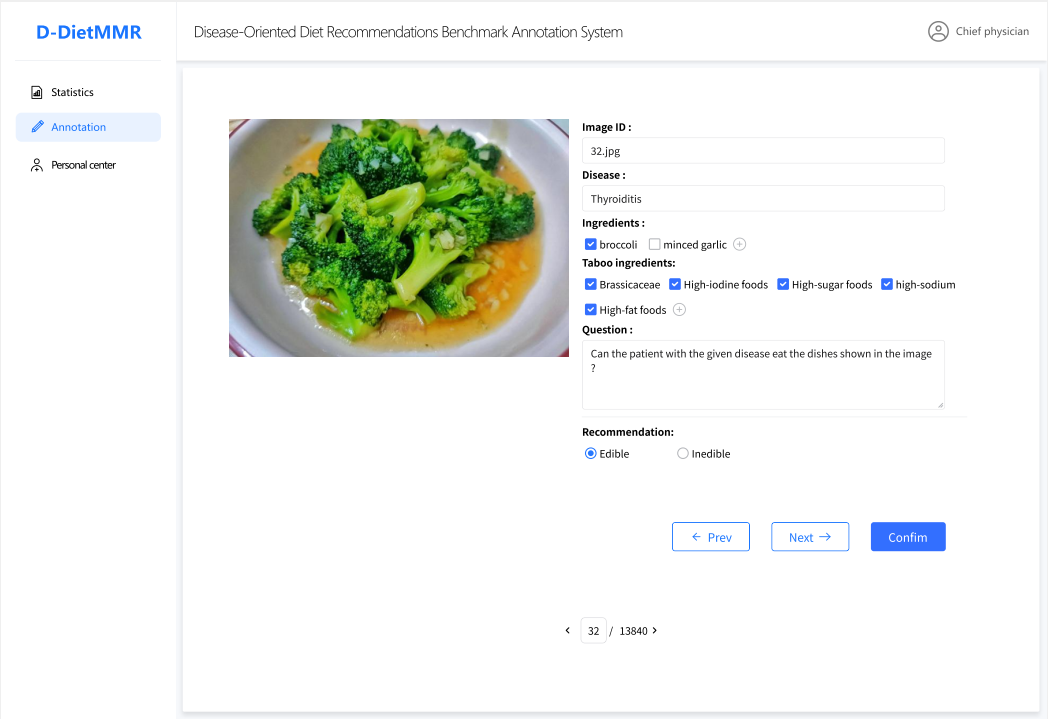

Figure A3: The reviewing page in the review annotation tool.

Table A1: Model comparison (Acc %) on the Top-5 dishes/diseases in D-DietMMR test set.

Models	Methods	Dishes					Disease				
		Sla.	Ste.	Don.	Sti.	Whi.	Rhi.	Dia.	Ora.	Adr.	Pne.
Closed-source LVLMs											
GPT-4o	ICL (one-shot)	71.82	42.59	73.02	81.40	47.06	67.24	70.83	63.16	76.79	65.45
GPT-4v	ICL (one-shot)	73.64	46.30	66.79	83.72	41.18	65.86	75.00	64.91	69.64	69.09
Qwen-VL-MAX	ICL (one-shot)	45.45	55.56	67.92	93.02	50.98	70.69	35.42	59.65	66.07	56.36
GLM-4V-Plus	ICL (one-shot)	72.46	35.19	64.31	71.43	69.23	55.17	70.21	56.14	70.91	65.45
Gemini-Pro-V	ICL (one-shot)	56.36	68.89	58.49	83.72	41.18	63.79	45.83	49.12	46.43	50.91
Open-source LVLMs											
LLaVA-7B	ICL (one-shot)	20.00	33.33	24.53	81.40	41.18	65.52	29.17	42.11	30.36	43.64
LLaVA-13B	ICL (one-shot)	71.82	22.22	73.58	34.88	58.82	34.48	70.83	57.89	69.64	56.36
Llama3.2-11B-V	ICL (one-shot)	69.32	45.12	48.32	75.44	49.56	66.67	55.32	58.44	49.22	35.32
Qwen2-VL-7B	ICL (one-shot)	50.91	70.37	56.60	86.05	56.86	58.62	60.42	59.65	46.43	67.27
Qwen2-VL-72B	ICL (one-shot)	43.64	74.07	54.72	86.05	47.06	65.86	39.58	61.40	53.57	49.09
InternVL2-8B	ICL (one-shot)	65.45	24.07	62.26	48.84	52.94	46.55	58.33	50.88	58.93	49.09
InternVL2-26B	ICL (one-shot)	41.82	55.56	62.26	60.47	49.02	53.45	52.08	61.40	50.00	52.73
InternVL2-40B	ICL (one-shot)	71.82	50.00	71.70	55.81	54.90	58.62	60.42	59.65	50.00	54.55
Medical Special LVLMs											
LLaVA-Med-7B	ICL (one-shot)	45.32	51.32	46.25	57.61	44.78	41.21	40.67	46.56	48.33	50.97
HealthGPT-M3	ICL (one-shot)	48.64	52.61	55.78	85.67	42.31	42.36	47.69	44.54	42.83	46.91
HuatuoGPT-V-7B	ICL (one-shot)	46.32	47.82	52.56	80.65	49.32	46.54	50.00	48.33	46.29	50.32
MedGemma-4B-IT	ICL (one-shot)	46.73	68.58	50.56	83.48	54.29	54.37	52.71	55.39	43.28	63.29
MedGemma-27B-IT	ICL (one-shot)	51.86	71.62	58.97	89.69	59.87	58.63	59.83	60.18	49.58	69.56
Other Methods											
CLIP	ICL (zero-shot)	68.18	67.04	71.70	81.40	68.63	63.79	62.50	57.89	67.86	52.73
BLIP2-Chinese	ICL (zero-shot)	50.00	51.85	47.17	76.74	45.10	62.07	45.83	52.63	53.57	43.64
Instruction-tuned LVLMs											
Qwen2-VL-7B-T (Ours)	ICL (one-shot)	80.23	77.13	74.37	93.26	78.52	77.13	79.42	80.97	88.78	90.11
Llama3.2-11B-V-T (Ours)	ICL (one-shot)	75.61	78.96	80.62	88.79	70.31	74.38	72.44	79.09	76.16	82.89

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The prompt for zero-shot direct prompting strategy

Role definition

You are a professional doctor.

Instruction

Your objective is to respond to the input question by answering either "yes" or "no".

Input

Question: "*Can the {disease} patient eat this dish shown is picture {id}?*"

Figure A4: An example of the zero-shot direct prompting strategy.

The prompt for one-shot prompting strategy

Role definition

You are a professional doctor.

Instruction

Your objective is to respond to the input question by answering either "yes" or "no".

Demonstrations

The following is an example of instruction to help you understand the task:
Question: "Can the hypothyroidism patient eat this dish shown in picture *1.jpg* ? "
Answer: "no"

Input

Question: "*Can the {disease} patient eat this dish shown is picture {id}?*"

Figure A5: An example of the one-shot prompting strategy.

The prompt for CoT prompting strategy

Role definition

You are a professional doctor.

Instruction

Your objective is to to reason out the final answer by answering the following questions:
1. You first answer the name of the dish shown in picture {id};
2. Identify the ingredients used in the dish;
3. List the taboo ingredients for patients with {disease};
4. Answer the question only "yes" or "no".

CoT Demonstrations

The following is an example of instruction to help you understand the task:
Question: "Can the adrenal cancer patient eat this dish shown in picture *17.jpg* ? "
1. The dish shown in picture 17.jpg is Fuqi Feipian.
2. The main ingredients of this dish include beef, beef tripe, beef tongue, peanuts, cilantro, chili oil, and sesame.
3. Patients with adrenal cancer should avoid foods that are high in fat, spicy or irritating, and high in cholesterol.
4. **Answer**: "no".

Input

Question: "*Can the {disease} patient eat this dish shown is picture {id}?*"

Figure A6: An example of the CoT prompting strategy.

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